Blueprint

The Pentagon, Machine-Checked: Abel's five-term relation for the dilogarithm in Lean 4

Special functions · the dilogarithm

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Statements

Theorem 1 — Dilog.rogersL_fiveterm 2

At a glance

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🚀 **Start here: Theorem 1**: Abel's five-term relation for the Rogers L-function on $(0,1)^2$.

Main results:

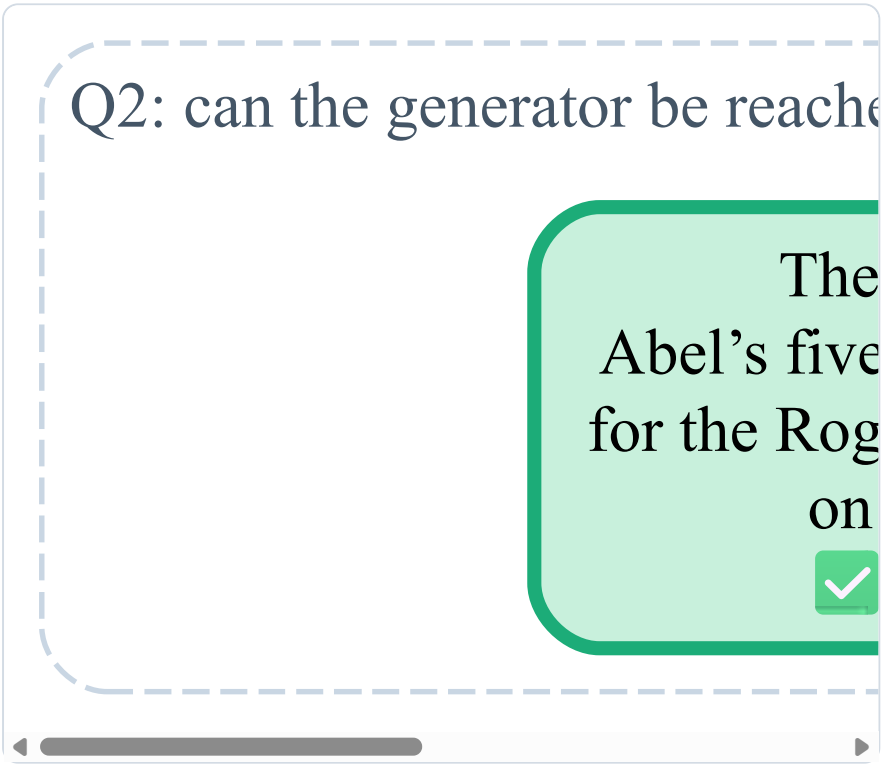
- **Theorem 1**: Abel's five-term relation for the Rogers L-function on $(0,1)^2$

1 results | ✅ **1** formalized in Lean | ◐ **0** with a part in Lean

💻 **0** checked by a program | written proof only | (a result can carry several tags)

🗺️ Map of the results (click a box to go to it)

Dependencies in the written proof: an arrow A → B means that this proof of B uses A. Every result rests on its written proof; the tags record what was checked besides, and each statement's "Evidence & scope" block says how far. The colour of a box is its strongest tag, in the order above; the tags in the box give the rest. Boxes with a thick border are the main results; click a box to read the statement.



1 Q2: can the generator be reached by the same three-move recipe?

Theorem 1. ✅ Lean (`Dilog.rogersL_fiveterm`)✓ 📕 p. 3

> *For all* $x, y \in (0,1)$,

$$\mathrm{L}(x)+\mathrm{L}(y)=\mathrm{L}(xy)+\mathrm{L}\Big(\tfrac{x(1-y)}{1-xy}\Big)+\mathrm{L}\Big(\tfrac{y(1-x)}{1-xy}\Big).$$

✅ Checked when this page was built: FiveTerm.lean compiles with Lean 4.35.0, no errors, no sorry. Axioms — rogersL_fiveterm, fiveterm_hasDerivAt_zero, hasDerivAt_rogersL, tendsto_rogersL_nhdsGT_zero, continuousAt_rogersL: propext, Classical.choice, Quot.sound. Reproduce: Lean 4.35.0, Mathlib f0b4c8150aa0; in that Lean project, `lake env lean FiveTerm.lean` (SHA-256 of the compiled file: `ce29fa490a366ddb…`; a source page on this site may show it with comments edited, under its own hash).

🆕 **Classical and new:** classical (Spence, Abel, Rogers); the paper claims the formalization: «to the best of my knowledge the first machine-checked five-term relation, on the first machine-checked dilogarithm, in any proof assistant» (§6), from a documented search of June 2026.

↳ Not used by another result of the paper.

Proof

Formalized in Lean (see the linked declarations).

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